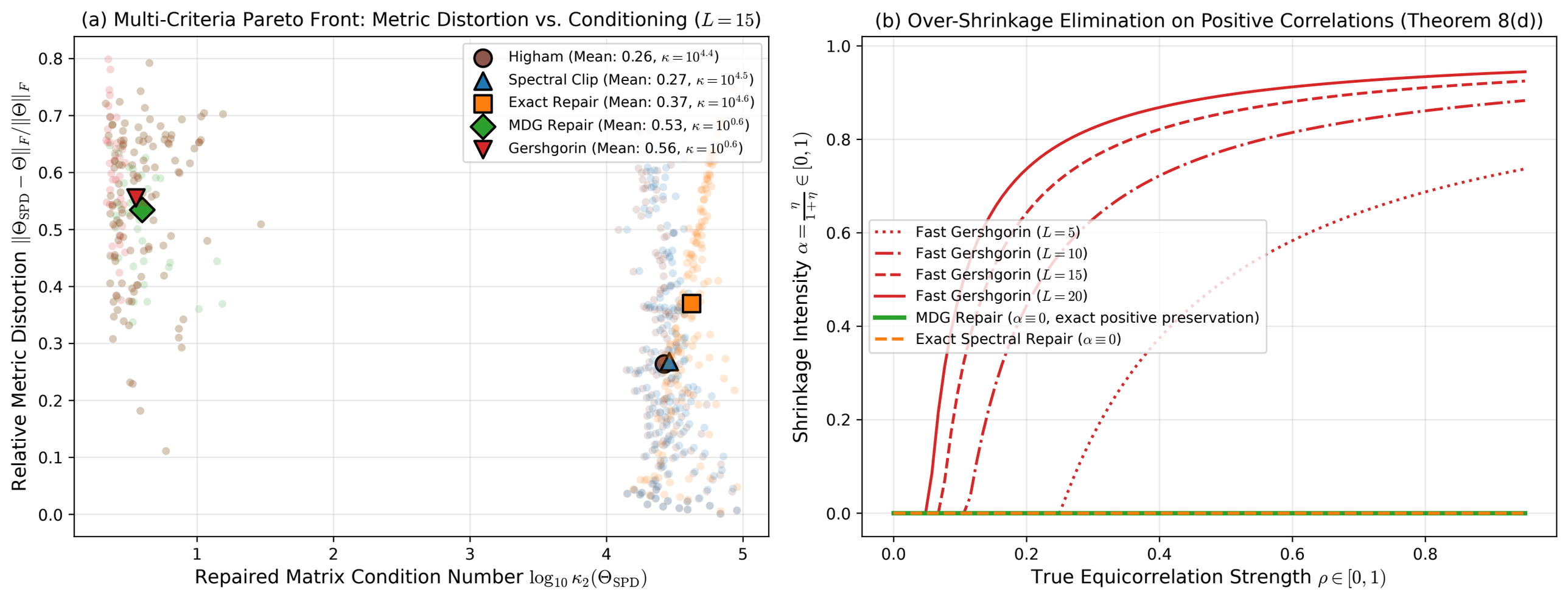
(a) Multi-Criteria Pareto Front: Metric Distortion vs. Conditioning ($L = 15$)
Relative Metric Distortion $\|\Theta_{\text{SPD}} - \Theta\|_F / \|\Theta\|_F$
Repaired Matrix Condition Number $\log_{10} \kappa_2(\Theta_{\text{SPD}})$
Higham (Mean: 0.26, $\kappa = 10^{4.4}$)
Spectral Clip (Mean: 0.27, $\kappa = 10^{4.5}$)
Exact Repair (Mean: 0.37, $\kappa = 10^{4.6}$)
MDG Repair (Mean: 0.53, $\kappa = 10^{0.6}$)
Gershgorin (Mean: 0.56, $\kappa = 10^{0.6}$)
(b) Over-Shrinkage Elimination on Positive Correlations (Theorem 8(d))
Shrinkage Intensity $\alpha = \frac{\eta}{1+\eta} \in [0, 1)$
True Equicorrelation Strength $\rho \in [0, 1)$
Fast Gershgorin ($L = 5$)
Fast Gershgorin ($L = 10$)
Fast Gershgorin ($L = 15$)
Fast Gershgorin ($L = 20$)
MDG Repair ($\alpha \equiv 0$, exact positive preservation)
Exact Spectral Repair ($\alpha \equiv 0$)